

END SEMESTER ASSESSMENT (ESA) - Jan - May 2024**UE22ME131B - Mechanical Engineering Sciences****Total Marks : 100.0**

1.a. The following observations were recorded during a trial of a four-stroke, single-cylinder oil engine.

Duration of trial = 30 minutes

Oil consumption = 4 litres (in 30 minutes)

Calorific value of the oil = 43 MJ/kg

Specific gravity of the fuel = 0.8

Average area of the indicator diagram = 8.5 cm^2

Length of the indicator diagram = 8.5 cm

Spring constant = 5.5 bar/cm

Brake load = 150 kg

Spring balance reading = 20 kg

Effective brake wheel diameter = 1.5 m

Speed = 200 rpm

Cylinder diameter = 30 cm

Stroke = 45 cm

Calculate (i) Indicated power (ii) Brake power (iii) Brake specific fuel consumption in kg/kWh and (iv) Indicated thermal efficiency. (8.0 Marks)

1.b. Explain the functions of the following components -

i) Convergent - divergent nozzle in a De Laval (impulse) steam turbine

(ii) Heat exchangers in a closed loop gas turbine

(iii) Spiral casing in a Francis turbine

(6.0 Marks)

1.c. (i) List any two differences between flat plate type solar collector and concentrating type solar collector.

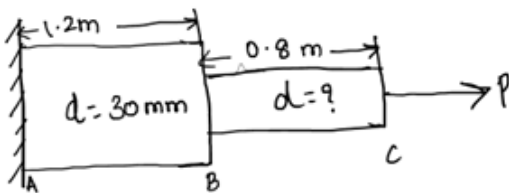
(ii) Describe the architecture of a parallel hybrid electric vehicle with a neat schematic diagram. (6.0 Marks)

- 2.a. (i) Differentiate between resilience and toughness.
(ii) Explain the behaviour of electro/magneto rheological fluids in the presence of the external field. (6.0 Marks)

- 2.b. Explain stress – strain diagram of a mild steel specimen subjected to tensile test, with a proper description of all salient points and regions of the diagram. (8.0 Marks)

- 2.c. A single axial load of magnitude $P = 58 \text{ kN}$ is applied at end C of the brass rod ABC. Knowing that $E = 105 \text{ GPa}$, determine the diameter 'd' of portion BC for which the deflection of point C will be 3 mm.

(6.0 Marks)



- 3.a. i) Define inversion of a mechanism.
ii) Describe with neat sketches, the inversions obtained when the cylinder and connecting rod respectively are fixed in a single slider crank chain mechanism. (8.0 Marks)

3.b. A flat belt runs on a pulley of 1m diameter and transmits 7.5 kW at a speed of 200 rpm. Taking angle of lap as 170° and coefficient of friction as 0.2, determine the necessary width of the belt if the ratio of maximum tension to width of the belt is not to exceed 196 N/cm. (6.0 Marks)

3.c. i) Explain reverted gear train.
ii) Determine the number of teeth for two toothed wheels to transmit a velocity ratio of $1/4$ between two parallel shafts, the centre of which are at a distance of 675 mm. Take module = 15 mm. (6.0 Marks)

4.a. Explain the following casting defects –
1) Hot tears
2) Cold shut
3) Blow holes (6.0 Marks)

4.b. Explain the working principle of oxy – acetylene gas welding with a neat sketch. Describe the three types of flames used in this welding. (8.0 Marks)

4.c. (i) List any two differences between hot working and cold working processes.
(ii) Explain the working principles of forging with neat sketch.

(6.0 Marks)

5.a. (i) What is the function of lead screw in a centre lathe machine?
(ii) Explain the working principle of taper turning by swivelling the compound rest with neat sketch.

(8.0 Marks)

5.b. Explain the working principles of counter boring and boring operations with neat sketches.

(6.0 Marks)

5.c. (i) A flexible automated system is capable of producing a variety of parts (or products) with virtually no lost production time. Justify the statement.
(ii) Explain open loop and closed loop control systems.

(6.0 Marks)

END SEMESTER ASSESSMENT (ESA) - JULY - 2023**UE22ME131B - Mechanical Engineering Sciences****Total Marks : 100.0**

1.a. Describe the way in which series hybrid electric vehicle and parallel hybrid electric vehicle are propelled, with corresponding block diagram representations.
(6.0 Marks)

1.b. The following particulars were obtained in a trial on a 4 stroke gas engine.
Duration of trial = 1 hour
No. of revolutions = 14000
Net brake load = 1470 N
Mean effective pressure = 7.5 bar
Gas consumption = 20000 litres
Calorific value of gas = 21 kJ/litre
Cylinder diameter = 250 mm
Stroke = 400 mm
Sum of diameters of brake drum and rope = 1.27 m
Calculate (i) Indicated Power (ii) Brake Power (iii) Indicated thermal efficiency (iv) Mechanical efficiency
(8.0 Marks)

1.c. Explain the functions of the following components –
(i) Draft tube in a Francis turbine
(ii) Convergent - divergent nozzle in a De Laval turbine
(iii) Heat exchangers in a closed loop gas turbine
(6.0 Marks)

2.a. (i) Spring steels are “resilient” while structural steels are “tough”. Bring out the basic difference between the highlighted attributes of the aforementioned materials.

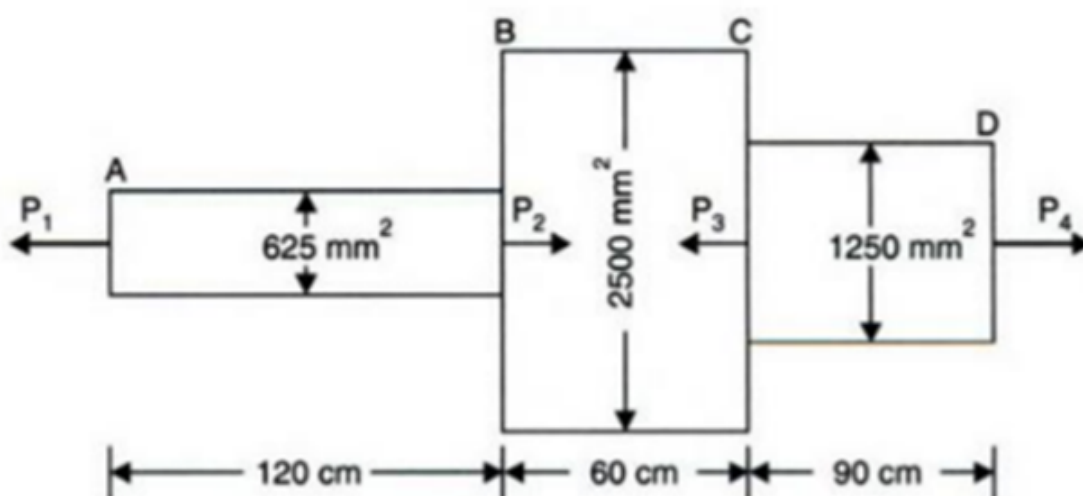
(ii) Explain the behaviour of electro/magneto rheological fluids in the presence of the external field. (6.0 Marks)

2.b. Explain stress – strain diagram of a mild steel specimen subjected to tensile test, with a proper description of all salient points and regions of the diagram.

(8.0 Marks)

2.c. A member ABCD is subjected to point loads $P_1 = 45$ kN, $P_2 = 365$ kN, $P_3 = 450$ kN and $P_4 = 130$ kN as shown in the following figure. Determine the total elongation of the member, assuming the modulus of elasticity to be 2.1×10^5 N/mm²

(6.0 Marks)



3.a. A flat belt runs on a pulley of 1m diameter and transmits 7.5 kW at a speed of 200 rpm. Taking angle of lap as 170° and coefficient of friction as 0.2, determine the necessary width of the belt if the ratio of maximum tension to width of the belt is not to exceed 196 N/cm. (6.0 Marks)

3.b. (i) Define (1) Successfully constrained motion (2) Rigid link
(ii) Identify and sketch the mechanism used for the following applications –
1) Coupled wheels of a locomotive in which the rotary motion of one wheel is transmitted to the other wheel.
2) Reciprocating engine. (6.0 Marks)

3.c. i) Explain briefly about double helical gears.
ii) Two spur gears A and B connect two parallel shafts that are 450 mm apart. Gear A runs at double the speed of gear B. Gear B runs at 150 rpm in the anti-clockwise direction. If the circular pitch is given to be 20, calculate the number of teeth on gears A and B. (8.0 Marks)

4.a. Explain the working principle of electric arc welding with a neat sketch. (6.0 Marks)

4.b. i) What do you mean by a Match plate pattern? Explain with a simple sketch.
ii) Explain briefly about Shrinkage Cavity defect. (6.0 Marks)

4.c. (i) List any two differences between hot working and cold working processes.
(ii) Explain the working principles of forging and drawing with neat sketches. (8.0 Marks)

5.a. i) Explain with neat sketches, the working principles of
a) Taper turning by swiveling the compound rest
b) Slab milling (6.0 Marks)

5.b. Define:
i) Range
ii) Resolution
iii) Precision
iv) Deadband (4.0 Marks)

5.c. Explain the following

i) Twisting joint

ii) Linear joint

iii) Programmable automation

(6.0 Marks)

5.d. Explain briefly the important application of industrial robots.

(4.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - JAN 2023

UE23ME131A - Mechanical Engineering Sciences

Total Marks : 100.0

1.a. The following are the details of a 4 stroke petrol engine.

Diameter of brake drum = 60.03 cm

Full brake load on drum = 250 N

Speed = 450 rpm

Calorific value of petrol = 40 MJ/kg

Brake thermal efficiency = 32%

Mechanical efficiency = 80%

Specific gravity of petrol = 0.82.

Determine (i) Brake power (ii) Indicated Power (iii) Fuel consumption in litres/second (iv) Indicated thermal efficiency (8.0 Marks)

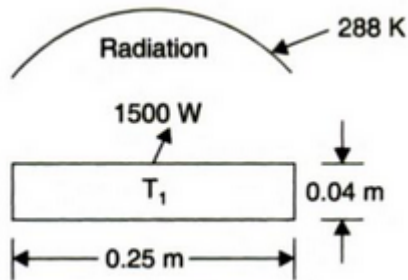
1.b. i) Describe the functions of the following components -

1) Jet deflector of a Pelton wheel

2) Draft tube of a Francis turbine (4 marks)

ii) Plug in hybrid electric vehicles often realize more savings in fuel costs than tradition hybrid electric vehicles. Justiy the statement. (2 marks)
(6.0 Marks)

1.c. An electric room heater (radiator) element is 25 cm long and 4 cm in diameter. The element dissipates heat to the surroundings at 1500 W mainly by radiation (neglect convection), the surrounding temperature being 15°C. Determine the equilibrium temperature of the element surface.



Note: Emissivity of the heater element = 1;

(5.0 Marks)

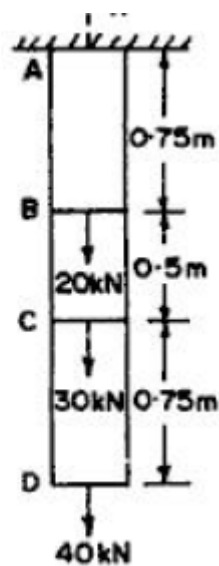
1.d. Explain the functions of absorber, heat exchanger and generator in a vapour absorption refrigeration system. (6.0 Marks)

2.a. i) List any two properties of a ceramic material with a corresponding application that utilises the property. (2 marks)

ii) Explain the principle of one way shape memory effect with a neat schematic diagram. (3 marks) (5.0 Marks)

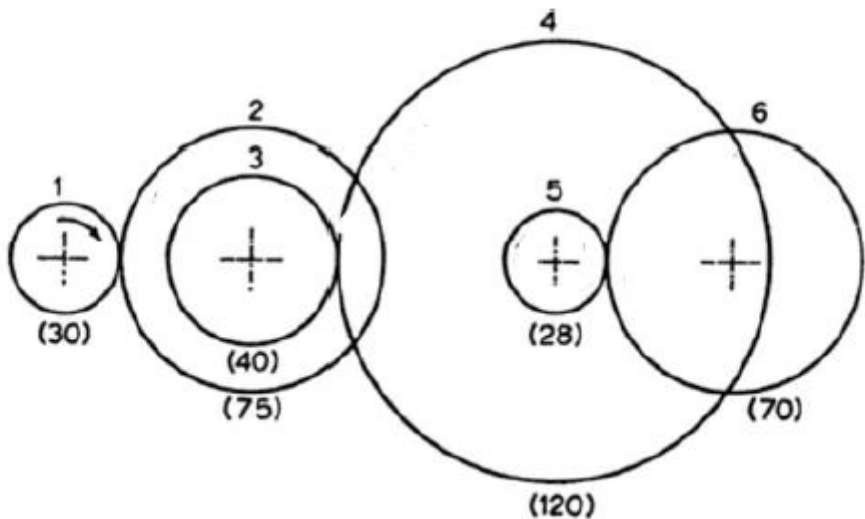
2.b. A steel bar of 2 m length and uniform section of 600 mm² is suspended vertically and loaded as shown in figure below. Taking $E = 2.05 \times 10^5$ N/mm², determine the total elongation of the bar.

(6.0 Marks)



- 2.c. i) Helical gears can be used at higher velocities than the spur gears and have greater load carrying capacity. Justify the statement. (2 marks)
- ii) In a compound gear train, shown in the following figure, the power is transmitted from a motor shaft to output shaft. The motor shaft is connected to gear 1 whereas the output shaft is connected to gear 6. The motor shaft is rotating at 1125 rpm in the clockwise direction. Determine the direction and speed of output shaft. The number of teeth on each gear are given below. (4 marks)

Gear	1	2	3	4	5	6
No. of teeth	30	75	40	120	28	70



(6.0 Marks)

2.d. A shaft rotating at 200 rpm drives another shaft at 300 rpm and transmits 6 kW through a belt. The distance between the shafts is 4 m. The smaller pulley is 0.5 m in diameter. Calculate the tension in the tight side of the belt, if it is an open belt drive. Take $\mu = 0.3$. (8.0 Marks)

3.a. i) What do you mean by a loose piece pattern? Explain with a simple sketch. (3 marks)

ii) Define Permeability of moulding sand with the description of any one casting defect that would be resulted by less permeability of sand. (3 marks) (6.0 Marks)

3.b. Explain the working principles of extrusion and sheet metal drawing with neat sketches. (8.0 Marks)

3.c. i) Describe any two functions of the coating in the coated consumable electrode used in electric arc welding. (2 marks)

ii) Explain the working principle of friction welding with a neat sketch. (4 marks) (6.0 Marks)

3.d. i) Differentiate between pitch and lead of a screw. (2 marks)

ii) Explain the working principle of rivetting with a neat sketch. (3 marks) (5.0 Marks)

4.a. i) The lead screw of a centre lathe is 6 mm. If a thread of 1.5 mm pitch is to be cut on a work piece, which is rotating at 120 rpm, what should be the speed of the lead screw? (2 marks)

ii) Explain the working principle of taper turning by swivelling the compound rest with a neat sketch. (6 marks) (8.0 Marks)

4.b. Explain the working principles of the following processes with neat sketches -

- 1) Counterboring (3 marks)
- 2) Side and Straddle Milling (3 marks)

(6.0 Marks)

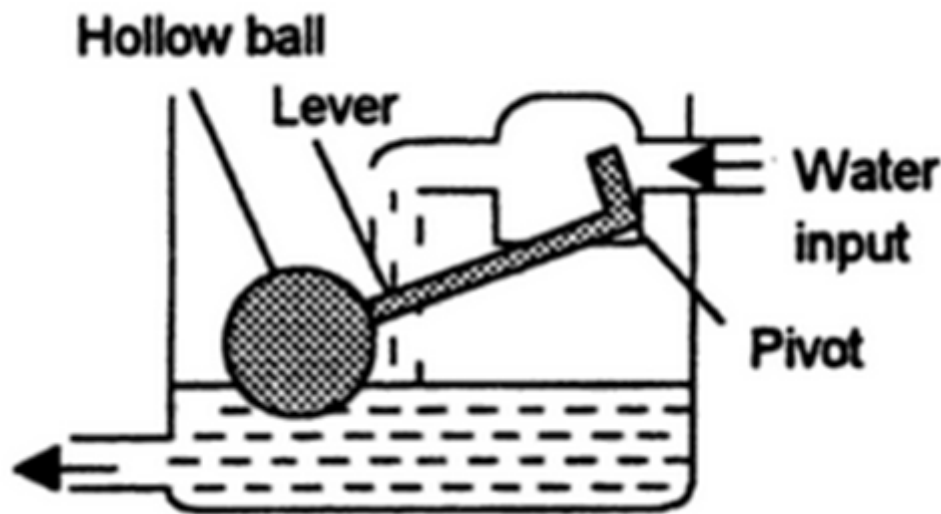
4.c. i) Explain the jointed arm and SCARA configurations of an industrial robot (Diagram not required). (4 marks)

ii) Pictorially depict the relative positions of the three types of automation for different production volumes and product varieties. (2 marks)

(6.0 Marks)

4.d.

i) The following figure represents a simple control system used to maintain a constant water level in a tank. The reference value is the initial setting of the lever arm arrangement so that it just cuts off the water supply at the required level. When water is drawn from the tank, the float moves downwards with the water level. This causes the lever arrangement to rotate and so allows water to enter the tank. This flow continues until the ball has risen to such a height that it has moved the lever arrangement to cut off the water supply. Identify the "**control unit**", "**measuring device**" and the "**process**" in the above mentioned control system. (3 marks)



ii) Describe the concept of IIOT based smart healthcare. (2 marks)
(5.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - JAN 2023

UE23ME131A - Mechanical Engineering Sciences

Total Marks : 100.0

1.a. A single-cylinder 4-stroke IC engine has a bore of 180mm, stroke of 200mm, and a rated speed of 300 rpm. Torque on the brake drum is 200 N-m and mean effective pressure is 6 bar. It consumes 4 kg of fuel in one hour. The calorific value of the fuel is 42000 KJ/Kg. Determine I) Brake Power II) Indicated Power III) Brake Thermal Efficiency IV) Mechanical Efficiency (8.0 Marks)

1.b. Explain the working principle of a Francis turbine. (4.0 Marks)

1.c. Illustrate the series & parallel architecture of the electric hybrid vehicle. (5.0 Marks)

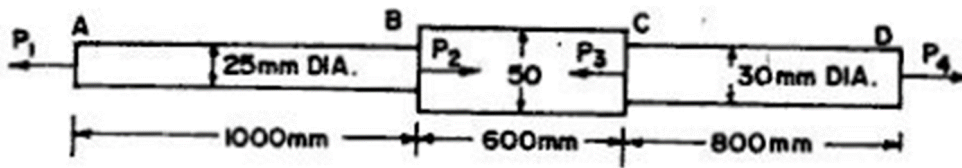
1.d. What is heat transfer and list the 3 modes of heat transfer? Explain the working principle of the vapor absorption refrigeration cycle. (8.0 Marks)

2.a. Compare Thermoplastics and Thermosetting plastics. (5.0 Marks)

2.b. Explain the shape memory alloys. (4.0 Marks)

2.c. A member ABCD is subjected to point loads P_1 , P_2 , P_3 , and P_4 as shown in the figure below. Calculate the force P_2 necessary for equilibrium if $P_1 = 10$ kN, $P_3 = 40$ kN, and $P_4 = 16$ kN. Taking the modulus of elasticity as 2.05×10^5 N/mm², determine the total elongation of the member.

(6.0 Marks)

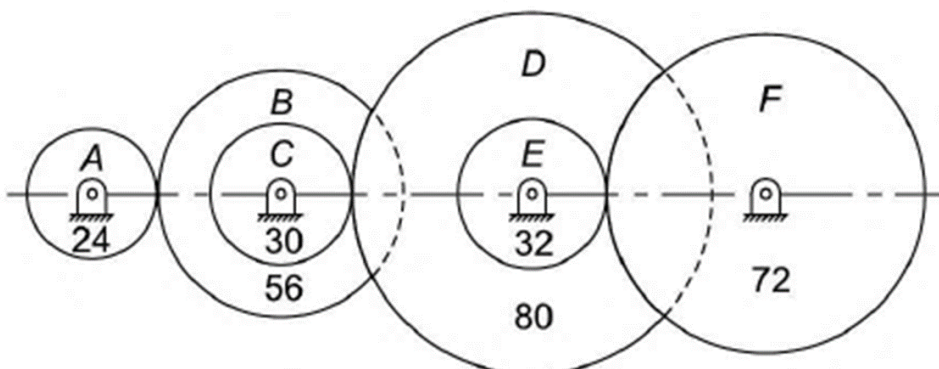


2.d. Two pulleys mounted on two parallel shafts that are 2 m apart are connected by a crossed belt drive. The diameters of the two pulleys are 500 mm and 240 mm. Determine the power transmitted if the larger pulley rotates at 180 rpm and the maximum permissible tension in the belt is 900 N. The coefficient of friction between the belt and pulley is 0.28.

(6.0 Marks)

2.e. A compound gear train shown in the figure consists of compound gears B – C and D – E. All gears are mounted on parallel shafts. The motor shaft rotating at 800 rpm is connected to the gear F. The number of teeth on gears A, B, C, D, E, and F are 24, 56, 30, 80, 32, and 72 respectively. Determine the speed of the gear F.

(4.0 Marks)



3.a. Define any 6 metal casting terms.

(6.0 Marks)

3.b. Explain loose piece patterns.

(6.0 Marks)

3.c. Explain extrusion and wire drawing related to the metal forming process.

(8.0 Marks)

3.d. Explain the working principle of resistance spot welding. (5.0 Marks)

4.a. Discuss the different taper-turning methods used in the lathe machine tool. (6.0 Marks)

4.b. Summarize the up-milling and down-milling operation. (5.0 Marks)

4.c. Explain the different types of mechanical joints used in industrial robots. (8.0 Marks)

4.d. Discuss Industrial Internet of Things (IIoT). (6.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - Jan-Feb

UE24ME141A - Mechanical Engineering Sciences

Total Marks : 100.0

1.a. i) Give thermodynamic definition of work transfer. (2 marks)

ii) The combustion gases of an I.C. engine expand within an enclosed piston and cylinder arrangement and follow the path $PV^{1.6} = C$. The pressure at the beginning of the power stroke is 5 MPa, and volume 50 cm³. At the end of the stroke the volume is 1500 cm³. Calculate the work developed during the power stroke. (4 marks) (6.0 Marks)

1.b. A thermocouple with test junction at $t^{\circ}\text{C}$ on a gas thermometer scale and reference junction at ice point is given by the relation

$$e = (0.25t - 5.1 \times 10^{-4} t^2) \text{ mV}$$

The millivoltmeter is calibrated at ice and steam points. If the gas thermometer reads 50°C, what will be the reading on this thermometer? (6.0 Marks)

1.c. i) Diesel engines are heavier than Petrol engines. Why? (2 marks)

ii) The following particulars were obtained in a trial on a 4 stroke gas engine.

Duration of trial = 1 hour

No. of revolutions = 14000 (in 1 hour)

Net brake load = 1470 N

Gas consumption = 20000 litres (in 1 hour)

Calorific value of gas = 21 kJ/litre

Sum of diameters of brake drum and rope = 1.27 m

Indicated Power = 28.63 kW

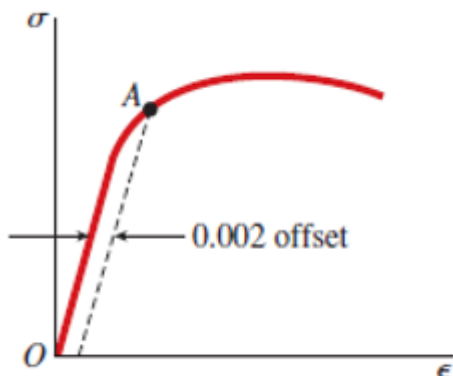
Calculate (i) Brake Power (ii) Indicated thermal efficiency (iii) Mechanical efficiency (6 marks) (8.0 Marks)

1.d. i) Define turbine efficiency, generator efficiency, and combined turbine-generator efficiency. (3 marks)

ii) Conceptually, series hybrid vehicle is an IC engine assisted electric vehicle. Substantiate the statement. (2 marks) (5.0 Marks)

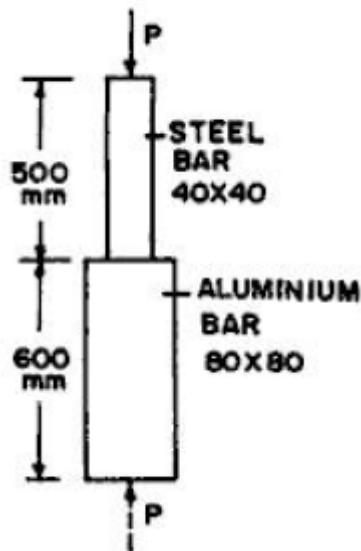
2.a. i) Give one example for a polymer matrix composite material with the proper identification of reinforcement and matrix phases. (2 marks)

ii) The following figure shows the stress strain diagram for an aluminium specimen subjected to uniaxial tensile test. Define point 'A' in the figure and its significance. (3 marks) (5.0 Marks)



2.b. A member formed by connecting a steel bar to an aluminium bar is shown in the following figure. Calculate the magnitude of force P , that will cause the total length of the member to decrease by 0.20 mm. The values of elastic modulus for steel and aluminium are 210 kN/mm^2 and 70 kN/mm^2 respectively.

(6.0 Marks)

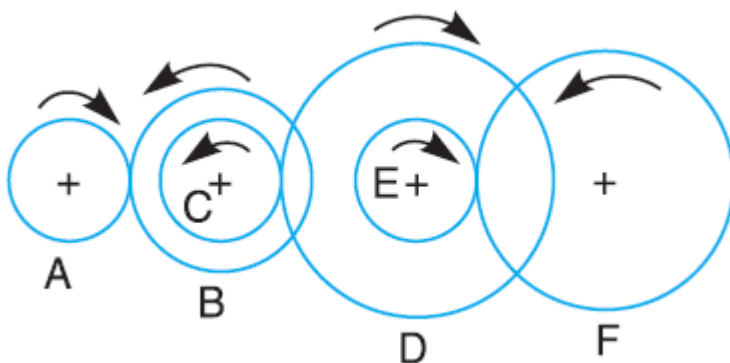


2.c. i) Axial thrust (force component acting along the axis of the shaft) is eliminated in case of double helical gears. Justify the statement. (2 marks)

ii) The gearing of a machine tool is shown in Fig. below. The motor shaft is connected to gear A and rotates at 975 rpm. The gear wheels B, C, D and E are fixed to parallel shafts rotating together. The final gear F is fixed on the output shaft. What is the speed of gear F? The number of teeth on each gear are as given below : (4 marks)

Gear	A	B	C	D	E	F
No. of teeth	20	50	25	75	26	65

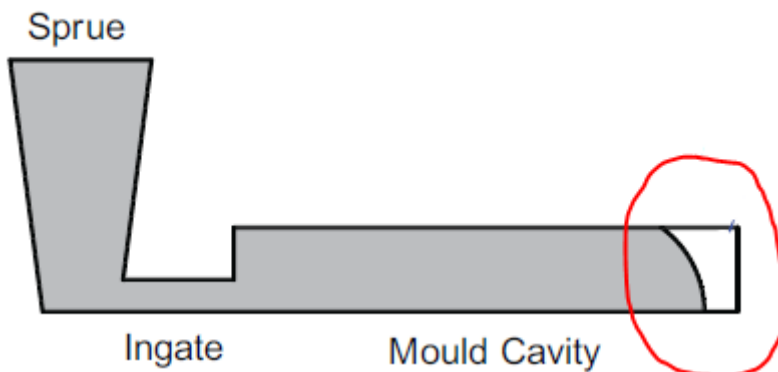
(6.0 Marks)



2.d. An open belt running over two pulleys of 24 cm and 60 cm diameters, connects two parallel shafts 3 m apart and transmits 3.75 kW from the smaller pulley that rotates at 300 rpm. The coefficient of friction between the belt and the pulleys is 0.3. Determine
 (i) tensions in the tight and slack sides of the belt
 ii) length of the belt required. (8.0 Marks)

3.a. i) Distinguish between liquid shrinkage and solid shrinkage as related to castings. Explain how these are taken care of in designing sand castings. (3 marks)

ii) The following figure highlights a particular type of casting defect. Identify and describe the defect with atleast one cause and one remedy for the same. (6.0 Marks)



3.b. i) A 40 mm thick plate is to be reduced to 30 mm in one pass in a rolling operation. Determine the minimum required coefficient of friction that would make this rolling operation possible, if the roll radius is 300 mm. (4 marks)

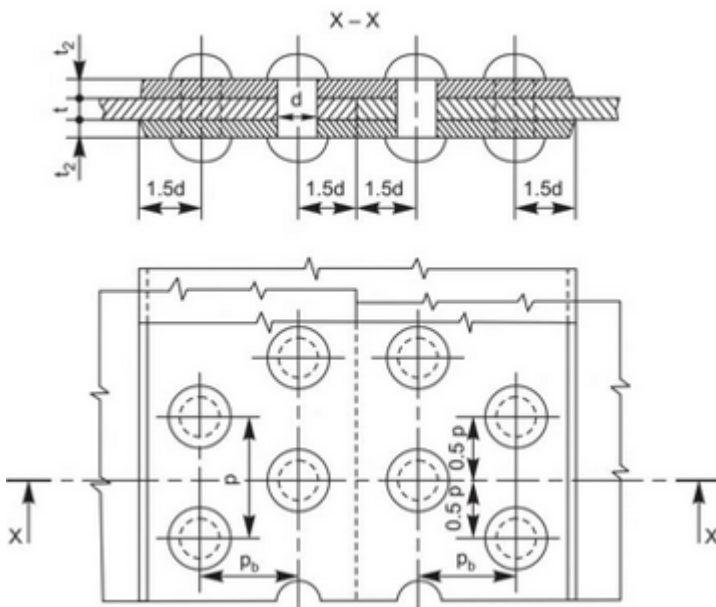
ii) Describe with a neat schematic sketch, the process of forward extrusion. (8.0 Marks)

3.c. Describe the working principle of oxy acetylene gas welding with a neat schematic sketch. (6.0 Marks)

3.d. i) Differentiate between major diameter and minor diameter of a thread. (2 marks)

ii) The following figure shows the front and top views of two plates connected through rivets. Identify the type of the rivetted joint. Mention all the attributes of the rivetted joint. (3 marks)

(5.0 Marks)



4.a. i) The following data is applicable for a turning operation. The length of job is 900 mm, diameter of job is 200 mm, feed rate is 0.25 mm/rev and optimum cutting speed is 300 m/min. Calculate the machining time in minutes. (4 marks)

ii) Explain the working principle of facing with a neat schematic sketch. (4 marks) (8.0 Marks)

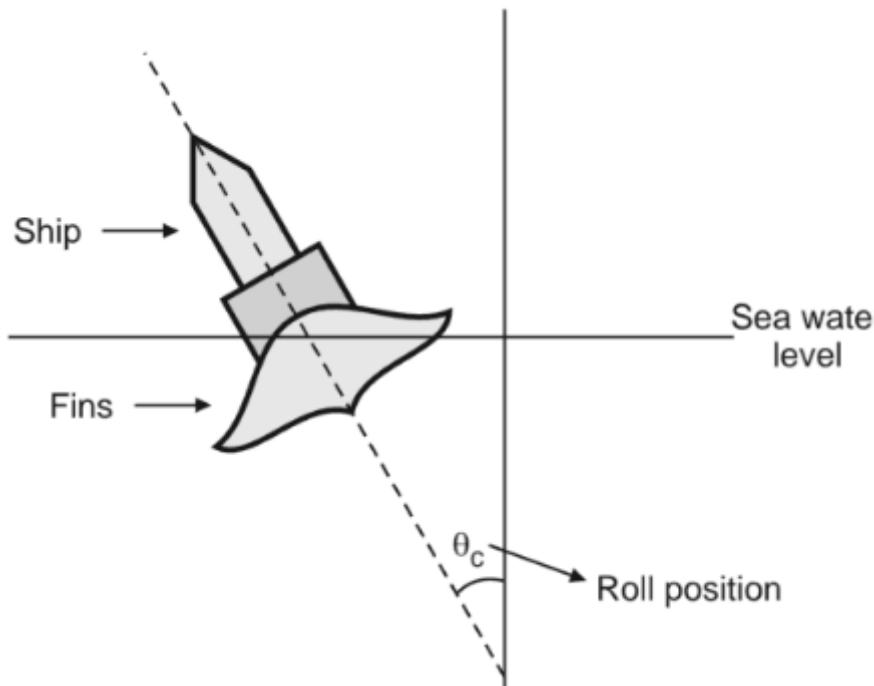
4.b. i) Differentiate between up milling and down milling. (3 marks)

ii) Explain the working principle of cylindrical grinding with a neat schematic sketch. (3 marks) (6.0 Marks)

4.c. i) A flexible automated system is capable of producing a variety of parts (or products) with virtually no lost production time. Justify the statement. (3 marks)

ii) In a ship stabilisation system, a roll sensor is used to measure the actual roll position θ_c which is compared with a desired roll position θ_r to generate a controlling signal that activates a fin actuator in proper way to stabilise the ship. Identify the "measuring device", "correction unit" and the "controlled variable" in this closed loop control system.

(6.0 Marks)



4.d. Define Material extrusion as a 3D printing process. List the steps of fused deposition modelling. (5.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - Jan-Feb

UE24ME141A - Mechanical Engineering Sciences

Total Marks : 100.0

1.a. Define the Zeroth Law of Thermodynamics and Illustrate the Zeroth Law of Thermodynamics as a basis for Temperature Measurement. (5.0 Marks)

1.b. A mass of gas is compressed in a quasi-static process from 80 kPa, 0.1 m³ to 0.4 MPa, 0.03 m³. Assuming that the pressure and volume are related by $p v^n = \text{constant}$, find the work done by the system. (5.0 Marks)

1.c. The following observations are taken during a trial on a four-stroke diesel engine.

Cylinder diameter = 25 cm

Stroke = 40 cm

Speed = 250 rpm

Brake load = 70 kg

Brake drum diameter = 2m

Diesel oil consumption = 0.1 liters/min

Specific gravity of fuel = 0.78

Calorific value of fuel = 43900 kJ/kg

Indicated Power = 24.54 kW

Determine (i) Brake Power (ii) Mechanical efficiency (iii) Brake thermal efficiency (iv) Indicated thermal efficiency. (8.0 Marks)

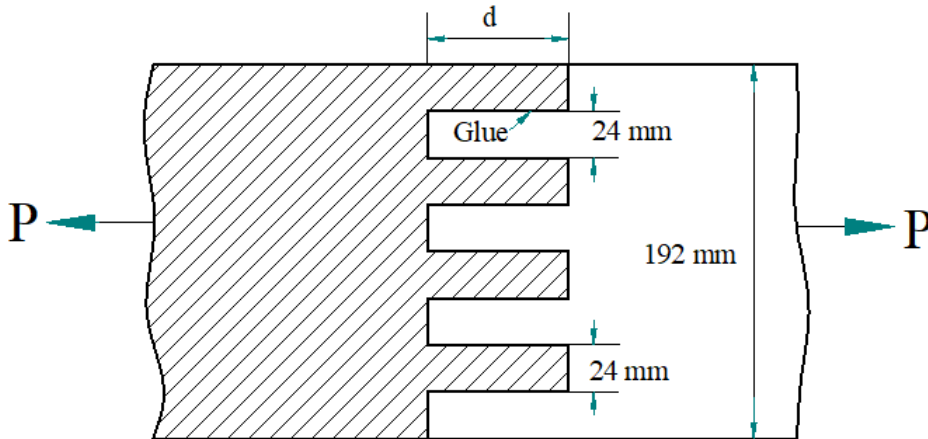
1.d. a) Compare any 3 performance characteristics of the 4 Stroke Petrol and Diesel Engine. (3 Marks)

b) Explain the parallel architecture of a hybrid Electric Vehicle. (4 Marks)
(7.0 Marks)

2.a. What are composite materials? Classify composite materials by the geometry of reinforcement and type of matrix, with an example. (6.0 Marks)

2.b. Two wooden planks, each 20 mm thick and 192 mm wide, are joined by the glued mortise joint shown. Knowing that the joint will fail when the average shearing stress in the glue reaches 800 KPa, determine the smallest allowable length d of the cut if the joint is to withstand an axial load of magnitude $P = 6$ kN

(4.0 Marks)



2.c. With the help of a stress-strain diagram for mild steel, explain the different stages of material behavior under tension. Discuss the key points such as **Yield point, Ultimate tensile strength, and fracture.**

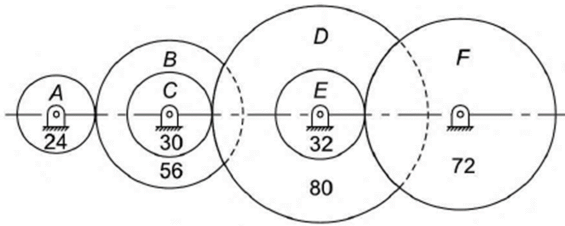
(6.0 Marks)

2.d. Two parallel shafts, connected by a crossed belt, are provided with pulleys 480 mm and 640 mm in diameter. The distance between the center lines of the shafts is 3 m. Determine by how much the length of the belt should be changed if it is desired to alter the direction of rotation of the driven shaft.

(6.0 Marks)

2.e. A compound gear train shown in the figure consists of compound gears B – C and D – E. All gears are mounted on parallel shafts. The motor shaft rotating at 800 rpm is connected to the gear A. The number of teeth on gears A, B, C, D, E, and F are 24, 56, 30, 80, 32, and 72 respectively. Determine the speed of the gear F.

(3.0 Marks)



3.a. Explain the Shrinkage and Machining Pattern allowance related to sand moulding.

(6.0 Marks)

3.b. Explain the forward and backward extrusion metal forming process.

(6.0 Marks)

3.c. Discuss Oxy-Acetylene welding and the chemical reaction of the combustion at the two stages. (8.0 Marks)

3.d. Define fasteners. Discuss the various types of fasteners commonly used in engineering applications, highlighting their functions and materials. (5.0 Marks)

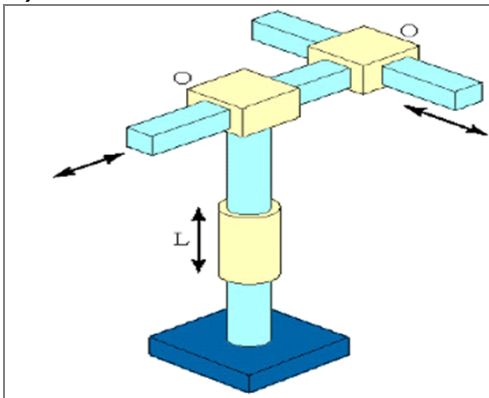
4.a. Discuss any four major operations that can be done using a Lathe Machine tool. (8.0 Marks)

4.b. Compare horizontal and vertical Milling machine tools. (5.0 Marks)

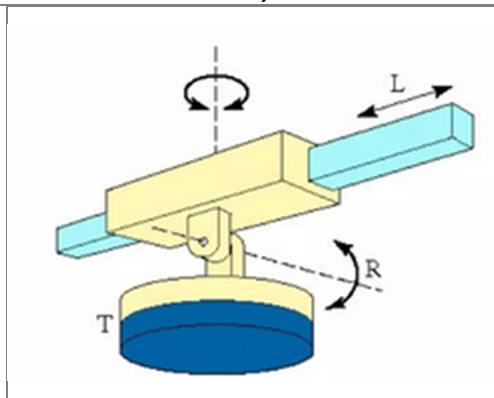
4.c. Explain the centreless grinding machine. (4.0 Marks)

4.d. Name and explain the below industrial robot configurations.

a)



b)



(4.0 Marks)

4.e. Briefly explain any four applications of the Industrial Internet of Things (IIoT). (4.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - Feb 2026

UE25ME141A - Mechanical Engineering Sciences

Total Marks : 100.0

1.a. Define a "Quasi-static Process" and explain its key features.
(4.0 Marks)

1.b. Explain the "Zeroth Law of Thermodynamics" and its practical application in temperature measurement.
(4.0 Marks)

1.c. A single-cylinder 4-stroke IC engine has a bore of 180mm, a stroke of 200mm, and a rated speed of 300 rpm. Torque on the brake drum is 200 N-m, and the mean effective pressure is 6 bar. It consumes 4 kg of fuel in one hour. The calorific value of the fuel is 42000 KJ/Kg. Determine: i) Brake Power
ii) Indicated power
iii) Brake thermal efficiency
iv) Mechanical efficiency
(8.0 Marks)

1.d. Explain the major components and working of a Battery Electric Vehicle (BEV) (6.0 Marks)

1.e. Define Hydraulic Machines and distinguish between Hydraulic Turbines and Pumps. (3.0 Marks)

2.a. What are smart materials? List any four examples of smart materials. Explain the working of one-way and two-way shape memory effects. (7.0 Marks)

2.b. Differentiate engineering stress and true stress. A brass rod in static equilibrium is subjected to axial load as shown in Figure a.

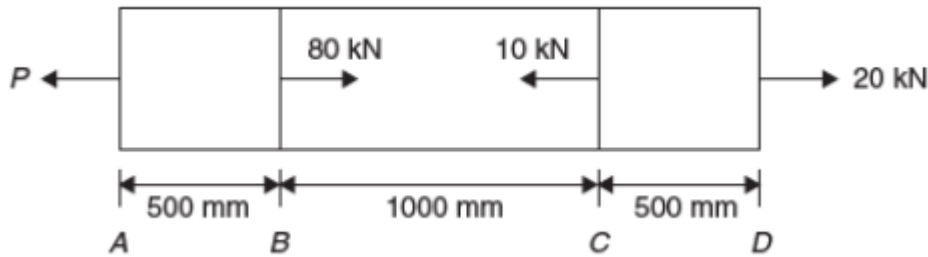
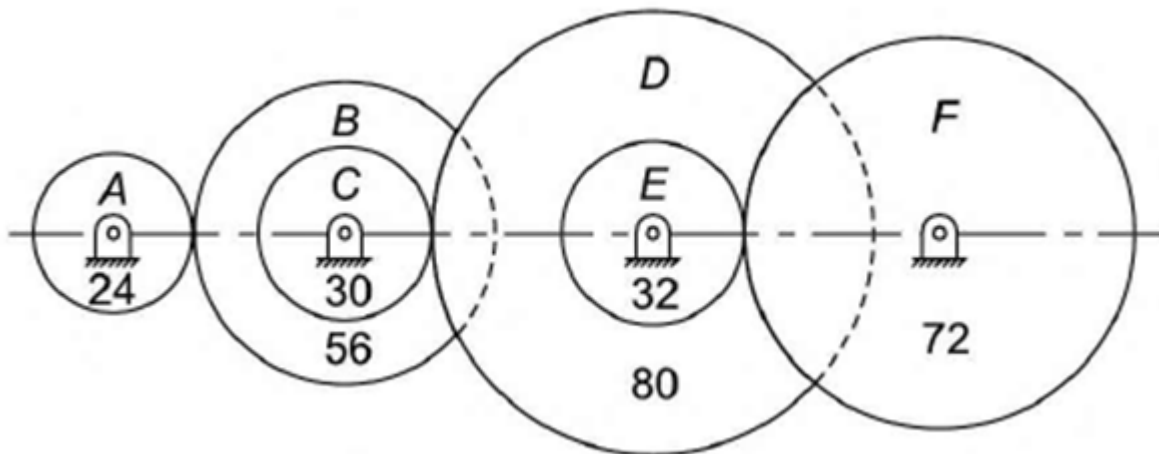


Figure a: A Bar Under Static Equilibrium

Find the load P and change in length of the rod if its diameter is 100 mm. Take $E = 80 \text{ GN/m}^2$. (10.0 Marks)

2.c. What is a gear train? Why are idler gears used when they do not affect velocity ratio?

A compound gear train shown in the figure consists of compound gears B – C and D – E. All gears are mounted on parallel shafts. The motor shaft rotating at 800 rpm is connected to the gear A. The number of teeth on gears A, B, C, D, E and F are 24, 56, 30, 80, 32 and 72 respectively. Determine the speed of the gear F.



(8.0 Marks)

3.a. Explain the process of soldering. List the commonly used fluxes in soldering and explain why flux is necessary. (7.0 Marks)

3.b. What do you understand by a pattern in casting? Explain the purpose of machining allowance and distortion allowance. (6.0 Marks)

3.c. Define metal forming. Explain the disadvantages of hot metal forming. With the help of neat sketches, describe the drawing-out and upsetting operations. (8.0 Marks)

3.d. Define oxy-acetylene welding? Explain the types of oxy-acetylene flames. (4.0 Marks)

4.a. What is milling? Differentiate between conventional and climb milling. (4.0 Marks)

4.b. Explain the following machines operations with neat sketch.

1. Turning
2. Knurling
3. Counter boring
4. Tapping
5. Counter sinking

(10.0 Marks)

4.c. What is a control system? Explain the basic elements of a closed loop control system with the help of a neat block diagram. (7.0 Marks)

4.d. Define the following:

a) Additive Manufacturing

b) Process Annealing

c) CNC Machining Centres

d) Roll in the wrist configuration of an industrial robot

(4.0 Marks)